User Acceptance Testing Form For

CSIS IT Ticket and Log System

Acceptance Form

PROJECT DETAILS

Project Name: *CSIS IT Ticket and Log System*

Project Manager:

Project Lead:

ACCEPTANCE DETAILS

Acceptance ID:

Requested By: *CSIS IT Ticket and Log System Development Team*

Date Requested:

Description:

The CSIS IT Ticket and Log System is a web-based application for the NWU Computer Science and Information Systems department. It replaces manual email-based IT reporting with structured fault submission, unique ticket tracking, technician job cards and start/stop time recording. Students and staff can follow requests and exchange comments with technicians. The system provides automated NWU email notifications, escalation records for Central IT, immutable audit history and management reporting. This form records acceptance against the functional and technical specifications. Results, remedial actions and customer approval are completed after testing.

ACCEPTANCE CRITERIA

Criteria 1:

Functional Requirements determined for the CSIS IT Ticket and Log System:

- NWU email and password authenticate users and route each role to its dashboard using a secure session; invalid credentials show a clear error. Reporter entry follows approved D01. **(F01)**
- Names, NWU email and assigned roles persist correctly; only authorized administrators can change permissions. **(F02)**
- The structured submission form captures Title, Issue Description, Category, Urgency, Location and Station ID using predefined category and urgency choices, and confirms successful submission. **(F03)**

Criteria 2:

Technical and Non-Functional Requirements determined for the CSIS IT Ticket and Log System:

- MongoDB, Express and Node.js implement persistence and backend logic. React SPA in FS versus Vanilla JavaScript in TS is resolved and evidenced under D02. **(T01)**
- Node.js is v20.x or greater; Express.js v4.x baseline is verified; target is Ubuntu Server 22.04 LTS or configured Windows Server with a Node process manager such as PM2. **(T02)**

- Clear validation prevents incomplete or invalid tickets; corrected input is accepted. Record the approved institutional email domains. **(F04)**
- Each ticket has a unique Ticket ID, creation time, correct reporter association and initial state; all fields persist in MongoDB. See D04. **(F05)**
- Priority is saved accurately and organizes the queue so higher urgency work is visible first; hazardous priority is tested separately in T cases. **(F06)**
- Each reporter sees their own requests and status history; another reporter ticket is inaccessible through both UI and API. **(F07)**
- Authorized technicians see the permitted ticket queue, filter by status, inspect history and sort by priority. Cards show Station ID, Priority and time since submission. Labels follow D04. **(F08)**
- The ticket records the technician and claim time, moves to the approved active state and updates the dashboard and history. **(F09)**
- Only approved transitions succeed; the ticket, queue and audit history remain consistent for New/Pending, Assigned, In Progress, Escalated and Resolved as agreed. **(F10)**
- Start and stop timestamps persist, active work changes the state to In Progress, duration is calculated to the second and the displayed total matches MongoDB. **(F11)**
- The total adds both sessions correctly in minutes/hours, persists after reload and has timestamped audit entries for the time records. **(F12)**
- Claim-to-resolution elapsed time is calculated and saved separately from accumulated active timer time and creation-to-closure turnaround, using approved D06 definitions. **(F13)**
- An empty summary prevents closure; valid resolution records notes and closure time, archives the ticket from active queues and triggers the user notification. **(F14)**
- Status becomes Escalated to NWU IT with reason and traceable administrative hand-off; integration records referrals without requiring write access to central IT databases. **(F15)**
- Messages remain linked to the correct Ticket ID, in time order, with sender identity and timestamp; both permitted parties see the same persisted conversation. **(F16)**
- Each new public comment triggers an alert to the other party with the relevant ticket link and without exposing another ticket conversation. **(F17)**
- Authorized IT staff can read the note; the student/staff reporter cannot view it in the UI, API responses or notification contents. **(F18)**

- The approved CSIS physical/virtual host supports the runtime and persistent ticket/audit storage; data location follows the agreed university boundary and D05. **(T03)**
- Functional parity is recorded for Chrome/Firefox/Edge, with FS Safari coverage; Windows 10/11, macOS and Linux browser access works without client installation. **(T04)**
- Mobile workflows match desktop functionality with readable responsive layouts and touch-friendly controls; FS Android 8.0+ and iOS 11.3+ support is documented or dispositioned. **(T05)**
- Core workflows remain usable on a 2 GB RAM dual-core desktop at 720p and a 2 GB quad-core mobile device with a 5-inch display, using keyboard/mouse or touch. Stable LAN, Wi-Fi or approved mobile access and 10/100 campus networking are supported. **(T06)**
- Purple/white branding and typography are consistent; deep purple #4B2C70 primary elements, slate secondary actions and labelled green/amber/red statuses follow the specified design. **(T07)**
- The specified WCAG 2.1 requirements, including contrast and keyboard access, are met; focus and labels support the workflows. Attach the audit scope and findings. **(T08)**
- Labels and errors are understandable; Log New Fault/recent-ticket access, station/priority/time details and lab/category analytics are present. Conceptual layouts are reviewed against approved designs. **(T09)**
- REST methods such as GET, POST and PUT exchange structured JSON between browser and backend; MongoDB storage and outbound Admin metrics follow the specified data flow. **(T10)**
- Web traffic uses HTTPS with TLS 1.2 or higher, protecting credentials and ticket data in transit; unencrypted access cannot expose protected data. **(T11)**
- Nodemailer sends notifications through the NWU gateway to @nwu.ac.za or @student.g.nwu.ac.za addresses. Port 587 with STARTTLS is the secure baseline; any port 25 use cited in TS p. 20 requires an approved encrypted relay configuration. **(T12)**

- Posted communication remains immutable and cannot be deleted; the complete troubleshooting history persists under the approved retention approach D08. **(F19)**
- An automatic receipt contains the unique Ticket ID and submitted summary, reaches the correct NWU inbox and links to the ticket. **(F20)**
- The NWU SMTP service dispatches correctly addressed, template-based updates for all specified events with the right status, dynamic data and direct ticket link. **(F21)**
- Administrators filter historical tickets by date range, technician and hardware/category. Only matching records are returned and supply consistent inputs to reports. **(F22)**
- Admin graphs and tables match the source data, show workload distribution and recurring faults, and use correct time definitions including creation-to-closure turnaround. **(F23)**
- Both report formats contain the selected data and accurate metrics; files are readable and accessible only to the authorized role. **(F24)**
- Critical actions have actor and timestamp records, every state change exists in audit_logs and each resolved ticket identifies its technician and closure time; records resist modification. **(F25)**
- Ticket lists, status and public messages update asynchronously without a full page refresh, with near real-time visibility and recorded latency. **(F26)**
- Both user groups complete the full reporting and communication workflow successfully and receive the final resolution update. **(F27)**
- All job card, notification, audit and reporting records agree for the same ticket, with role boundaries maintained throughout. **(F28)**

Supporting Project Plan Requirements:

- Automated API tests have a 100% pass rate and there are zero critical security vulnerabilities; attach the release-specific reports. **(P01)**
- The project-plan sub-2.0-second response target is assessed at the 100-user profile; retain this evidence separately from the TS minimum of 50 connections. **(P02)**
- The resolution email dispatches within 2 minutes of the trigger. Record inbox arrival separately so dispatch and delivery are distinguishable. **(P03)**
- The dedicated environment and email relay are provisioned and verified; unresolved dependencies have owners and cannot be recorded as passed without evidence. **(P04)**
- Email processing does not block application requests and sustained test traffic leaves no unresolved notification backlog. **(T13)**
- Signed JWTs establish identity and role for protected requests; invalid or absent tokens cannot claim, close or alter tickets or audit data. D01 governs reporter entry. **(T14)**
- Management authentication operates using the internal module; no unavailable central Active Directory/SSO API is required for the initial release. Future integrations remain separately scoped. **(T15)**
- Reporters are limited to their own tickets; technicians manage approved workloads; reports/system logs and permission changes are restricted to authorized administrators. Resolve queue scope in D04. **(T16)**
- Stored passwords are salted bcrypt hashes with the configured high work factor; no plaintext credential storage is present. Record configuration evidence without copying secrets. **(T17)**
- Server validation using express-validator prevents NoSQL/injection attacks, XSS script execution and unauthorized database access, including when client validation is bypassed. **(T18)**
- Only the application server IP can connect to MongoDB; direct client or unapproved-host access is denied. **(T19)**
- Personal and operational records have documented encryption and secure handling controls; names and NWU emails support incident resolution. Attach POPIA and university policy review evidence for acceptance. **(T20)**
- Records are archived or anonymized after the academic cycle according to the approved policy, while audit traceability and immutable operational history are handled as agreed in D08. **(T21)**
- Standard input responses and ticket acknowledgement are within 2.0 seconds; record timings, network conditions and any failures. PP page-load checks are also captured. **(T22)**

- The CSIS Network Administrator receives repository ownership, VM/SSH and MongoDB administrative access through secure channels, environment/IP/connection documentation and user guides. Sponsor reviews deployment and handover evidence. **(P05)**
- A support owner, clear user guide and escalation hand-off exist; the planned automatic legacy-mailbox reply directs users to the portal and outstanding project issues have recorded actions. **(P06)**

Optional and Future Requirements:

- The design can be extended to screenshot/log uploads; implemented upload functionality is a future enhancement unless explicitly included in this release. **(O01)**
- Offline access is optional/future scope; record implementation evidence or an approved N/A disposition without making it a current mandatory gate. **(O02)**
- The optional printed output is readable through standard OS/browser printing; otherwise record approved N/A. **(O03)**
- Future Outlook/calendar and full central-system integration remain separately approved work; initial escalation record and local management login remain testable without them. **(O04)**

- At least 50 concurrent connections are supported without data corruption or service decline. Capture response times, error counts and evidence that the notification queue drains. **(T23)**
- The system remains usable without client slowdown or significant lag. Record observed limits and agree a dataset and qualitative acceptance decision where no numeric budget is specified. **(T24)**
- Polling or supported WebSockets promptly refresh both views; one consistent technician assignment is retained and conflicting claims do not overwrite each other silently. **(T25)**
- Measure against the FS target of 99.9% during standard academic hours; record window, downtime and result separately from the technical target until D03 is approved. **(T26)**
- Measure both TS requirements: 98% uptime in that window and no more than 9.5 hours unplanned downtime per semester. Record period and resolve the differing targets under D03. **(T27)**
- All 10 students submit successfully without verbal help and each submission takes less than 60 seconds; attach participant timing evidence. **(T28)**
- Physical-risk tickets receive Critical/Hazardous priority and a prominent red labelled alert for immediate attention; no actual hazardous equipment is needed for testing. **(T29)**
- Technicians and management can identify repeated location-specific failures for investigation, including potential overheating/cooling patterns. **(T30)**
- Transactional handling restores the last known good state or a fully committed update; no partial or lost hazard ticket remains. **(T31)**
- At the predetermined interval the Lab Manager receives the escalation notification. Thirty minutes is an example in TS, not a fixed threshold until approved. **(T32)**

Baseline Decisions Before Sign-off:

- D01: Confirm mandatory reporter login versus NWU-email submission without account creation.
- D02: Confirm React MERN frontend versus the Vanilla JavaScript frontend stated in the technical specification.
- D03: Approve the uptime target, operating calendar and measurement period; both source targets remain listed.
- D04: Agree New/Pending and Assigned/In Progress transitions, escalation labels and technician queue permissions.
- D05: Confirm internal hosting and the university data-location boundary versus the cloud option mentioned in TS.
- D06: Keep active work time, claim-to-resolution elapsed time and creation-to-closure turnaround distinct.
- D07: Record both the 50-connection technical minimum and the 100-user project-plan performance profile.
- D08: Agree how post-2026 archiving or anonymization preserves approved immutable audit history.
- D09: Approve Critical/Hazardous priority, the hazard escalation timeout and Lab Manager recipient.

- Hardware-fault submission displays approved NWU laboratory safety reminders, including power-down guidance. Digital records show reported hazards and actions for institutional safety review and OHSA evidence. **(T33)**
- MongoDB backups run automatically and successfully at least once per 24-hour window; evidence identifies the latest recoverable backup. **(T34)**
- The database returns to its last known good backed-up state within 2 hours, with ticket, user, time-log and audit integrity verified. **(T35)**
- Every endpoint has JSDoc or Swagger documentation; ESLint or equivalent reports no critical style violations. snake_case data names and PascalCase modules follow TS conventions or documented exceptions. **(T36)**
- The maintainer can run the application in less than 20 minutes using the README; commented modular code supports future maintenance and deployment to other departments. **(T37)**
- A downloadable PDF quick-start guide, login-page FAQ and user/technician/Admin manuals cover access/login and password recovery, submission, categories, ticket status, timers, secure comments, reports, user roles and troubleshooting. **(T38)**
- Architecture, ERD, state transitions, wireframes, schemas, server deployment, SMTP setup and failure troubleshooting are available, current and professionally typed; external references follow NWU Harvard style. **(T39)**
- Source and technical documents are version controlled, secure SSH/HTTPS supports repository access and the documented update process delivers the approved release to the departmental server. **(T40)**
- The release uses approved free/open-source tooling and university resources with zero new licensing costs; initial launch is restricted to CSIS labs/offices within the 2026 scope. **(T41)**

ACCEPTANCE RESULTS

Testing methods used:

Reviewer for undertaking the UAT: _____

Date the test was completed: _____

Overall results: *Not assessed*

Total: _____ ***Passed:*** _____ ***Failed:*** _____ ***Blocked / Not run:*** _____

Remedial Action Required:

Acceptance	Reviewer	Date	Result
Criteria 1: Functional Specification. Refer to requirements F01 to F28 stated above.			
Criteria 2: Technical Specification. Refer to requirements T01 to T41 stated above.			

CUSTOMER APPROVAL

Supporting Documentation:

CSIS IT Ticket Log System Functional Specification v1.0.2
CSIS IT Ticket Log System Technical Specification v1.0.1
IT Ticket and Log System Project Plan and completed UAT evidence

Signature: _____ **Date:** _____

SUPPORTING ACCEPTANCE RECORD

Project Team: Shani Nezar and Lia Theunissen

Project Sponsor: Selouna Visser

Release or Build: _____

Environment and URL: _____

Acceptance Decision: *Not assessed*

Evidence and defect record location:

Conditions of acceptance and approved baseline decisions:

Execution and review:

Use an agreed test environment and record actual results and evidence against the F, T, P and O identifiers. F01-F28 are functional criteria; T01-T41 are technical criteria; P01-P06 are supporting project-plan checks; O01-O04 are optional or future scope. Record Pass, Fail, Blocked or approved N/A for each executed check in the linked evidence record. Summarize the outcomes on the acceptance results page. Leave unexecuted checks as Not run. Technical reviewers supply API, security, performance and recovery evidence. Record defect owners, remedial actions and retest outcomes before customer approval.

Source references:

Functional Specification v1.0.2, cover dated 11 May 2026, 16 PDF pages. Technical Specification v1.0.1, dated 9 May 2026, 28 PDF pages. Project Plan DOCX revision 1.2, dated 28 May 2026, and the supplied Project Plan PDF. The acceptance criteria consolidate repeated statements while preserving the requirement scope. The HartSpec form supplies the visual layout; its completed project results are not carried over.